

# The Mole Concept

Moles, molar mass and Avogadro's number

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

**DATA**

H = 1.008   C = 12.01   N = 14.01   O = 16.00   Na = 22.99   Mg = 24.31   Cl = 35.45

Ca = 40.08

Avogadro constant:  $6.022 \times 10^{23} \text{ mol}^{-1}$ . Show your working and give answers to an appropriate number of significant figures.

## A. Understanding the mole

1. What is a mole? Answer in one sentence.

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2. One mole of carbon has a mass of 12.01 g, but one mole of magnesium has a mass of 24.31 g. Do the two samples contain the same number of atoms? Explain.

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3. A student says “one mole of oxygen gas contains  $6.022 \times 10^{23}$  oxygen atoms.” What is wrong with this statement?

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## B. Molar mass

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4. Calculate the molar mass of each substance.

(a) methane,  $\text{CH}_4$  (b) ammonia,  $\text{NH}_3$  (c) calcium carbonate,  $\text{CaCO}_3$

5. Calculate the molar mass of (a) glucose,  $\text{C}_6\text{H}_{12}\text{O}_6$ , and (b) magnesium chloride,  $\text{MgCl}_2$ .

## C. Converting between mass, moles and particles

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6. How many moles are in 25.0 g of methane,  $\text{CH}_4$ ?

7. What is the mass of 0.150 mol of calcium carbonate,  $\text{CaCO}_3$ ?

8. How many molecules are in 0.500 mol of ammonia,  $\text{NH}_3$ ?

9. How many moles of water are in a sample containing  $1.50 \times 10^{24}$  water molecules?

10. What is the mass of  $3.01 \times 10^{23}$  atoms of sodium?

#### D. Counting atoms inside compounds

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11. How many moles of hydrogen atoms are in 2.00 mol of glucose,  $C_6H_{12}O_6$ ?

12. How many oxygen atoms are in 10.0 g of carbon dioxide,  $CO_2$ ?

13. What mass of chlorine is contained in 1.00 mol of magnesium chloride,  $MgCl_2$ ?

#### E. Challenge

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14. Which contains more molecules: 10.0 g of water,  $H_2O$ , or 10.0 g of carbon dioxide,  $CO_2$ ? Show your reasoning.

15. A 5.00 g sample of a metallic element contains 0.125 mol of atoms. Calculate the molar mass of the element and use the data box to identify it.